

DEPARTMENT OF PHARMACOLOGY
DIKIOHS
DOW UNIVERSITY OF HEALTH SCIENCES

PROFESSOR & HEAD OF DEPARTMENT
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SENIOR LECTURER: Dr. Sana
LECTURER: Dr. Anam
Dr. Hafsa

INTRODUCTION OF THE SUBJECT

THEORY

- **DEFINITIONS/TERMINOLOGY**
- **SOURCES & CLASSIFICATION**
- **ROUTES OF DRUG ADMINISTRATION**
- **PHARMACOKINETICS**
absorption, distribution, metabolism, excretion
- **PHARMACODYNAMICS**
receptor types, mechanism of action, dose response curves,
adverse drug reactions, drug-drug interactions, uses & contra-
indications

SYSTEMIC PHARMACOLOGY

- ANS
- CNS
- DIURETICS & CVS
- BLOOD
- GIT
- RESPIRATORY SYSTEM
- ENDOCRINE SYSTEM
- PAIN MANAGEMENT
- CHEMOTHERAPY

TEACHING METHODS

LECTURES

- **Power Point presentations**
- **questions on previous lecture at the start of the next one**

PRESENTATIONS (if scheduled)

RECOMMENDED TEXT BOOK:

PHARMACOLOGY; LIPPINCOTT'S ILLUSTRATED REVIEW, 10th (or latest) edition

REFERENCE BOOK:

BASIC & CLINICAL PHARMACOLOGY; KATZUNG, 14th (or latest) edition

ROUTES OF DRUG ADMINISTRATION

Department of Pharmacology

LEARNING OBJECTIVES

After the completion of this lecture students of 1st year BDS will be able to :

- Classify the routes of drug administration.
- Understand the specific advantages and disadvantages of oral route of drug administration.
- Understand the specific advantages and disadvantages of parental route of drug administration.

Route of administration: **path** by which a drug is introduced into the body.

Choice of an appropriate route depends on:

PATIENT

Clinical condition: Unconscious/ Fits / Vomiting

Nature of illness: acute, chronic

Site of desired action: Localized/approachable or generalized /not approachable

DRUG

Physical/Chemical properties of drug: Solid/Liquid/Gas, Solubility, Stability, pH...

Rate & extent of absorption of drugs from different routes

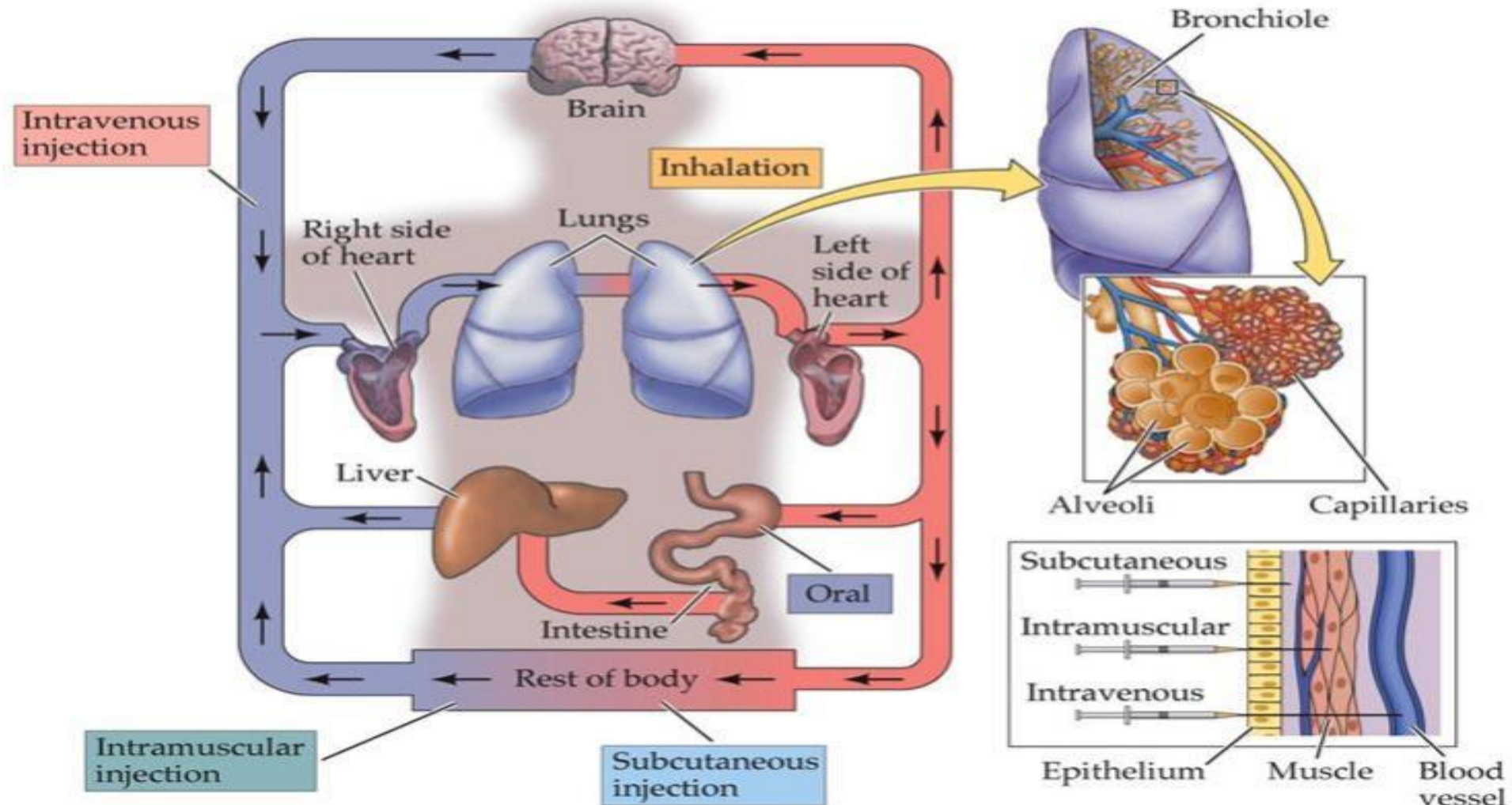
Effect of gastric acid/ digestive enzymes/1st pass metabolism

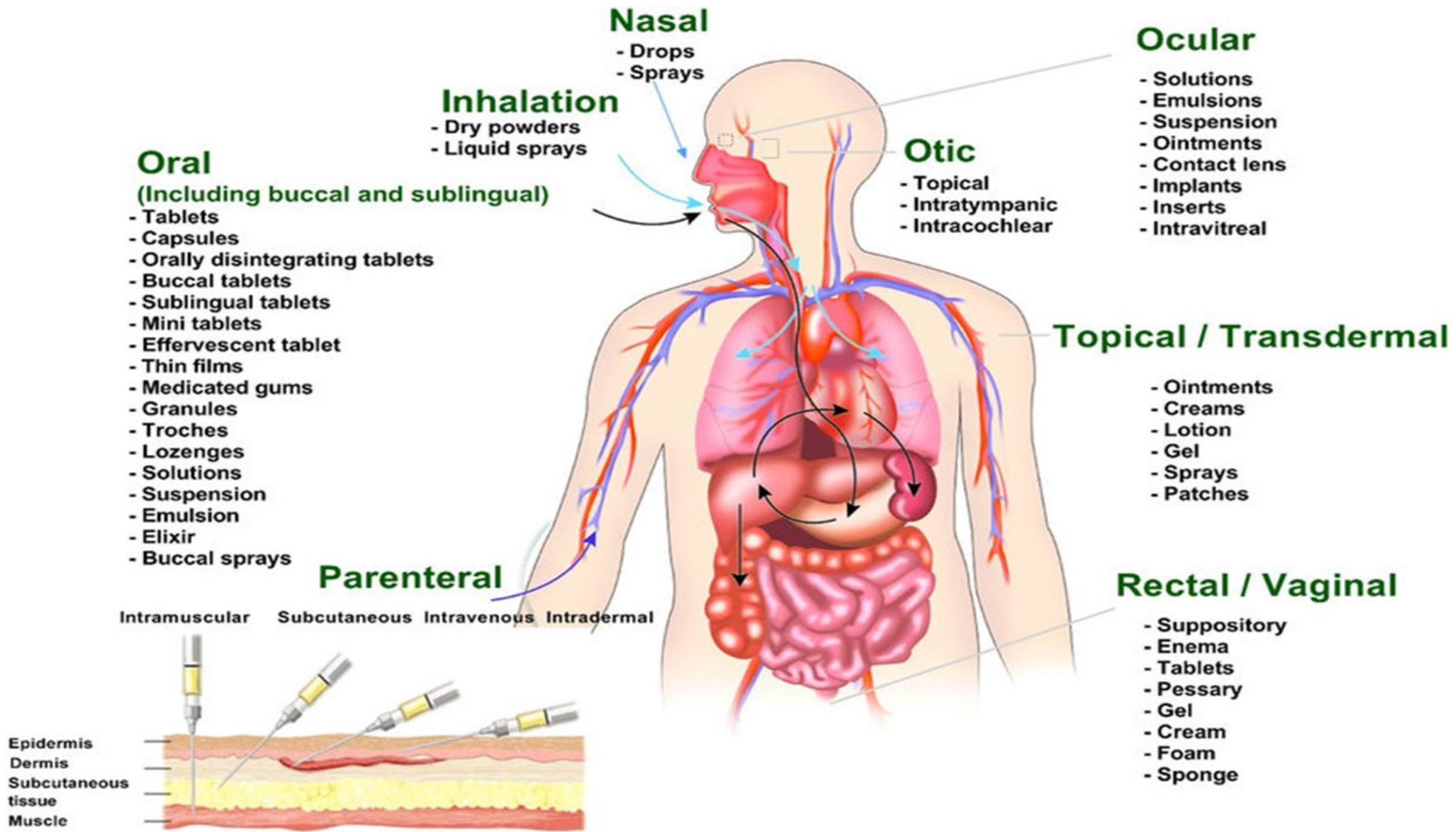
Rapidity of onset of action

ROUTES OF DRUG ADMINISTRATION

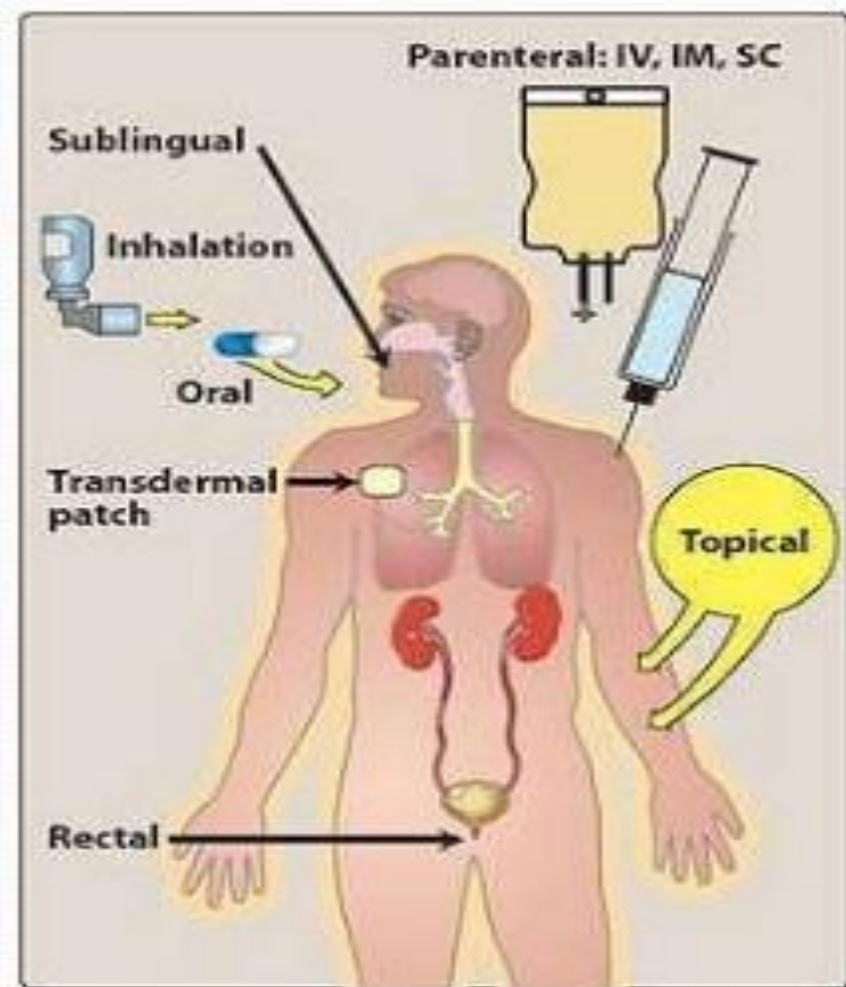
- 1) Enteral
- 2) Parenteral
- 3) Inhalation
- 4) Topical
- 5) Transdermal
- 6) Rectal
- 7) Intrathecal/Intraventricular
- 8) Intra-articular etc.

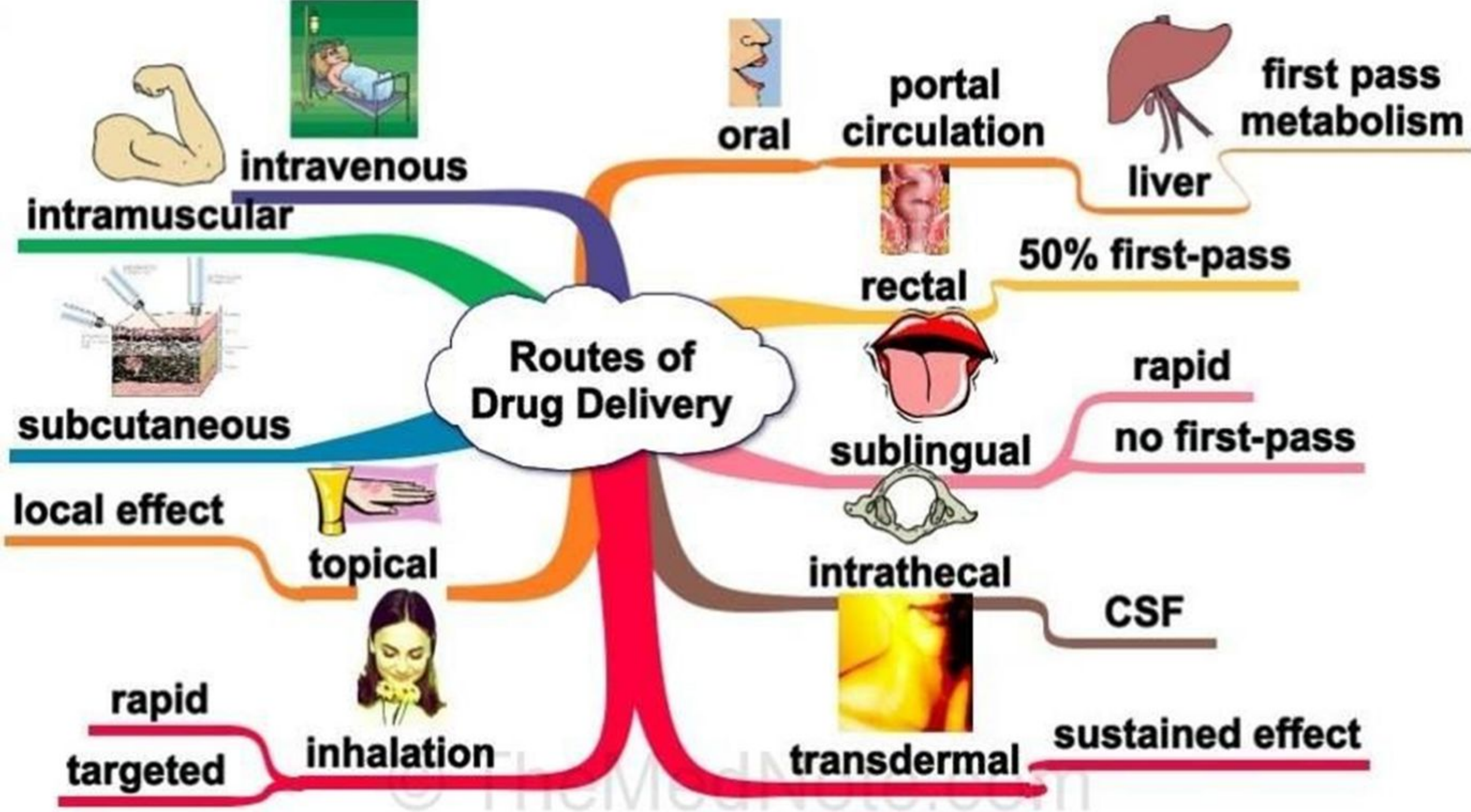
Routes of Drug Administration





- Enteral; oral, sub-lingual (buccal), rectal. Note soluble, enteric coated or slow release formulations
- Parenteral; iv, im, sc, id, it, etc. Different rates of absorption, different plasma peaks. Note iv infusors
- Skin; for local or systemic effect - note patches
- Lungs; inhalation; local or systemic effect?
- Vaginal; (usually local)
- Eye; (usually local)





ENTERAL ROUTE

ENTERAL

- ORAL
- SUBLINGUAL
- BUCCAL (BETWEEN CHEEK AND GUM)



Oral administration



Sublingual administration is where the dosage form is placed under the tongue



Buccal administration is where the dosage form is placed between gums and inner lining of the cheek

ENTERAL

- Enteral administration, or administering a **drug by mouth**, is the **safest** and most **common**, **convenient**, and **economical** method of drug administration.
- When the drug is given in the mouth, it may be swallowed, allowing oral delivery, or it may be placed under the **tongue (sublingual)**, facilitating direct absorption into bloodstream.

ADVANTAGES OF ORAL ROUTE

e.g. Tab. Acetaminophen

- Most common
- Safest
- Economical
- Convenient method of drug administration
- Reversible as drug toxicity or overdose may be overcome by giving an ANTIDOTE
- Lower risk of systemic infection

DISADVANTAGES OF ORAL ROUTE

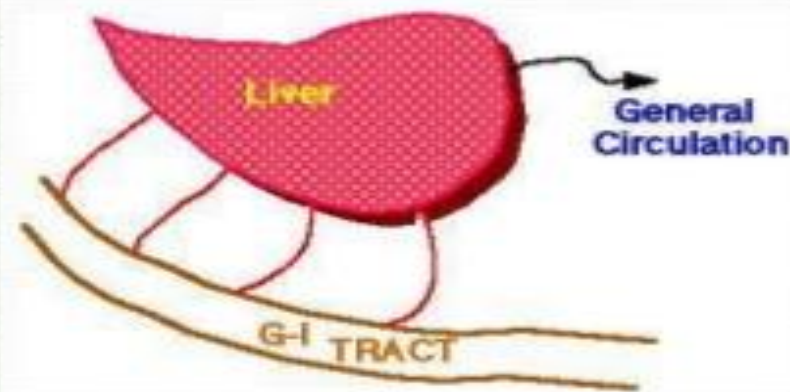
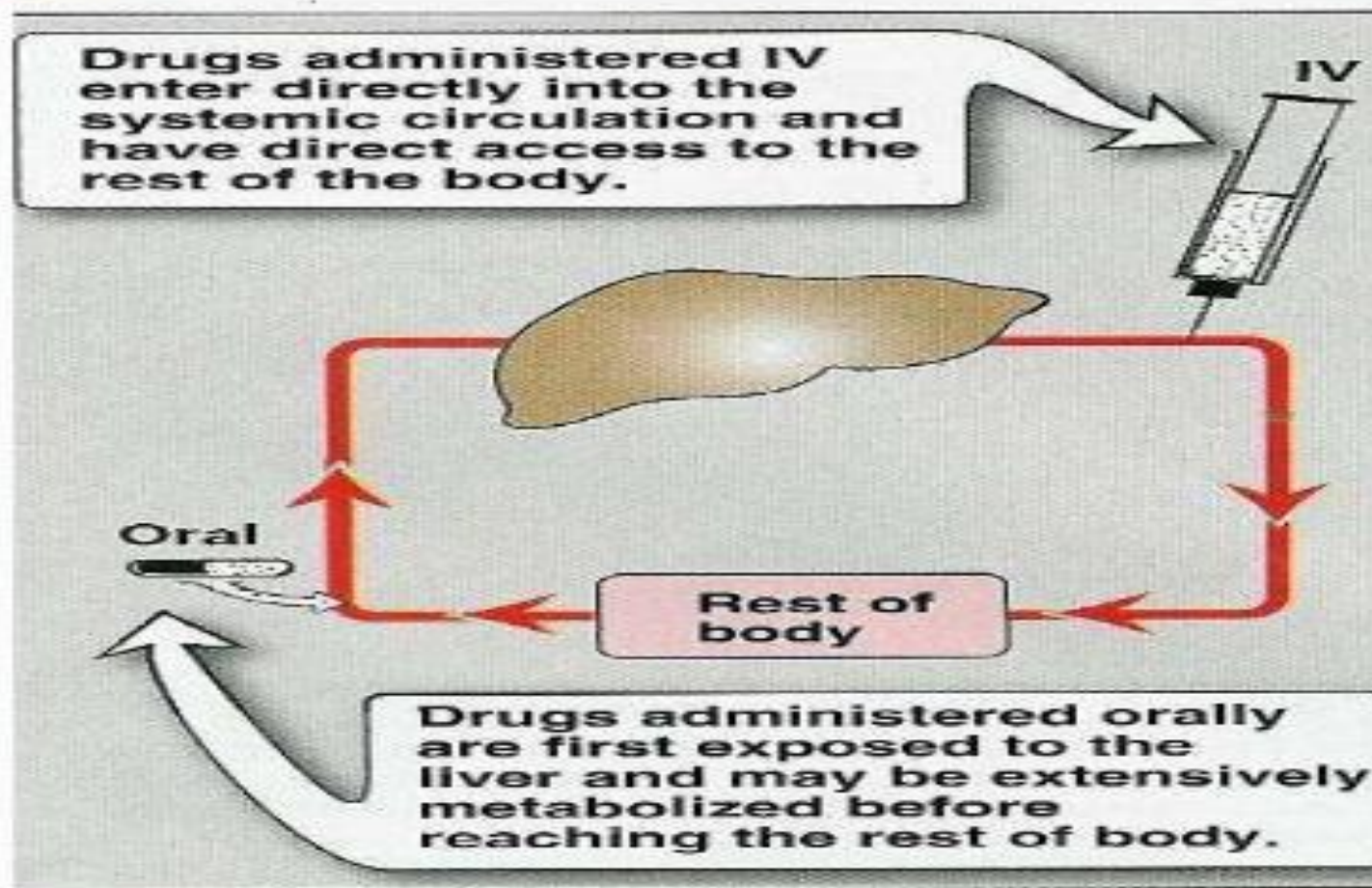
- Due to first pass - metabolism, drug bioavailability is low
- Has to face **harsh** acidic stomach environment
- Slow absorption, so not suitable for use in emergencies
- May cause GIT disturbances e.g. nausea/vomiting/irritation

ENTERIC COATED PREPARATION:

- Wide range of oral preparation is available including **enteric coated** and **extended release** preparations.
- **ENTERIC COATED PREPARATION:**
- Chemical envelope that protects the drug from stomach acid, delivering it instead to the less acidic intestine, where coating dissolves and releases the drug.
- Enteric coating is useful for certain drugs(OMEPRAZOLE, ASPIRIN)

EXTENDED RELEASE PREPARATIONS

- They have **special coatings** or ingredients that control drug release, thereby allowing for **slower absorption** and **prolonged duration of action**.
- They are dosed less frequently.
- They are advantageous for drugs with short half lives. For e.g. Half life of morphine is 2 to 4 hours and it must be administered six times daily for continuous pain relief.
- However , only two doses are needed when extended release tablets are used.



First-pass metabolism can occur with orally administered drugs.

SUBLINGUAL/BUCCAL ROUTE

- drug is placed under tongue or crushed in mouth and spread over the buccal mucosa. E.g. GTN
- **ADVANTAGES**
 - • Drug absorption is quick
 - • Quick termination
 - • First-pass avoided
 - • Can be self administered
 - • Economical

PARENTERAL ROUTE

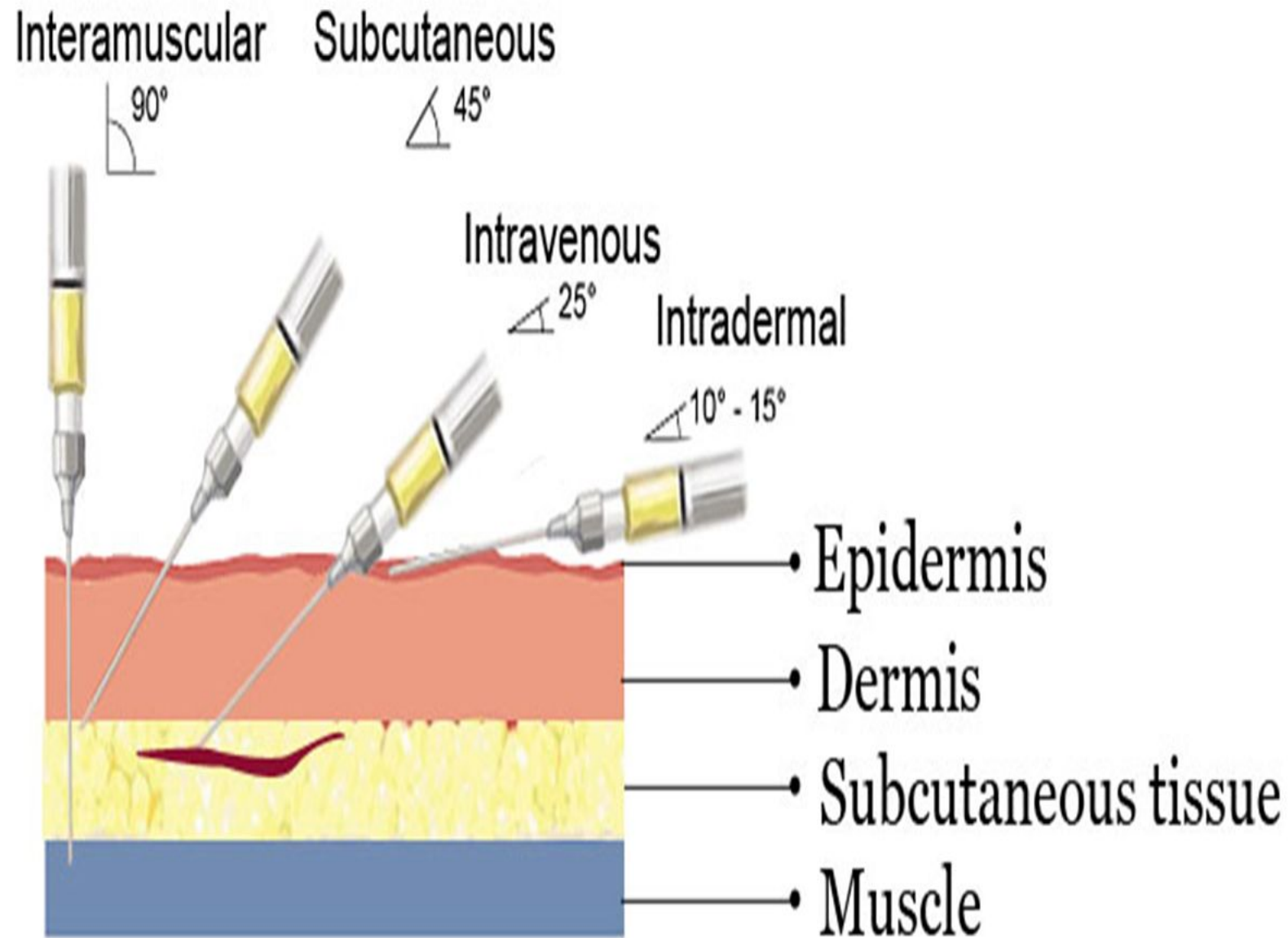
PARENTERAL

- Direct delivery of drug in to systemic circulation without intestinal mucosa ,
- Parenteral route is used for drugs that are poorly absorbed from GI tract
(Heparin) or unstable in GI tract (Insulin).
- Used for unconscious patients.

PARENTERAL ROUTES

CLASSIFICATION:

- INTRA-VENOUS
- INTRA-MUSCULAR
- SUBCUTANEOUS
- INTRA-THECAL
- INTRA-ARTICULAR
- INTRA-OSSEOUS
- INTRA-ARTERIAL
- INTRA-DERMAL



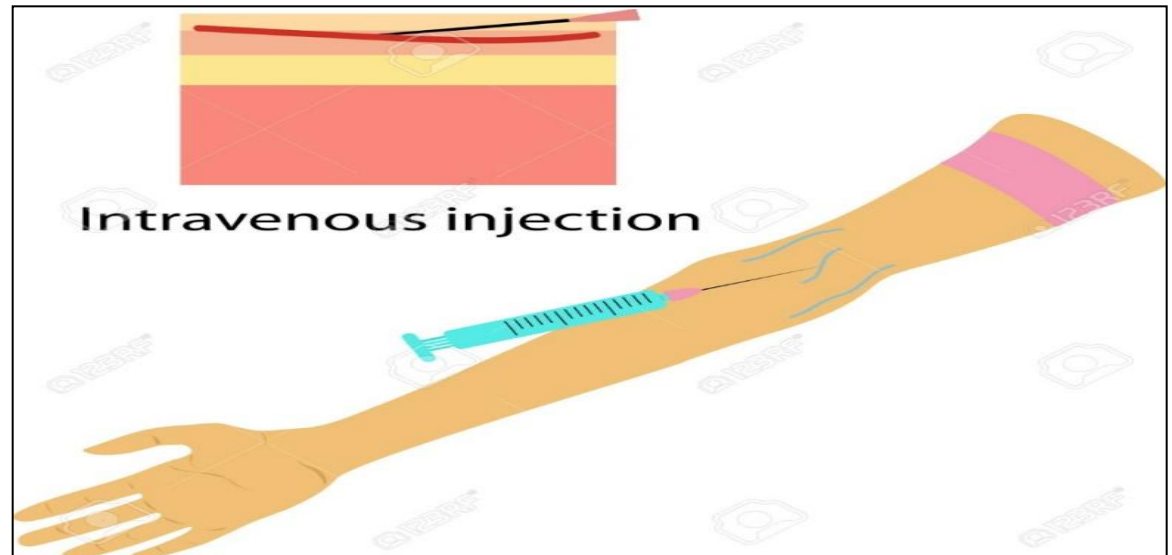
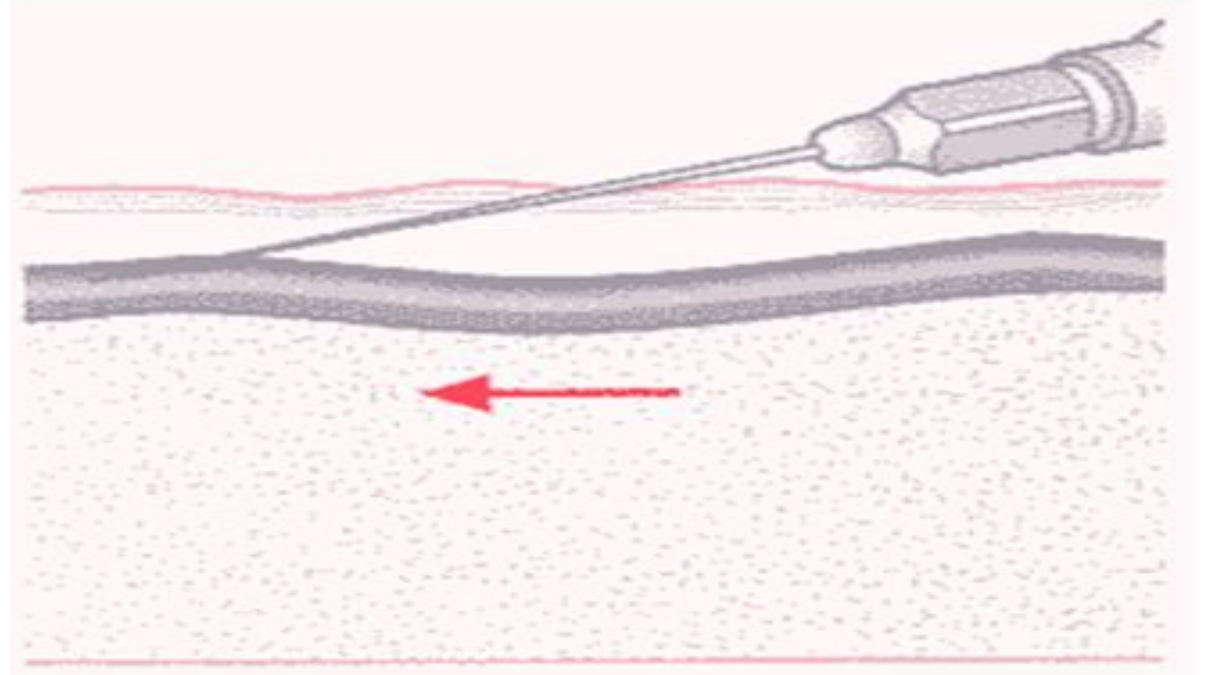
MAJOR PARENTERAL ROUTES

- **Intravascular**
- **Intramuscular (I.M.)**
- **Subcutaneous (S.C.)** (into subcutaneous tissue)
- **Intradermal**

ADVANTAGES OF INTRAVENOUS ROUTE

e.g. Inj. Tramadol

1. Bioavailability is 100%
2. Rapid response---- in Emergencies
3. Ideal route for the drugs :
 - a. having high molecular weight
 - b. to be given in large volume
 - c. that are irritant in nature



DISADVANTAGES OF INTRAVENOUS ROUTE

1. Risk of Local irritation /tissue damage at the site of injection
2. Strict aseptic technique are needed as I/V injection may introduce pathogens directly into the blood stream
3. Risk of Hemolysis/ Thrombosis
4. Risk of thrombophlebitis
5. Adverse drug reaction may occur after rapid delivery of high conc .of drug. It would be very difficult to manage & sometime it may be irreversible



INTRAMUSCULAR ROUTE

e.g. Inj.Diclofenic sodium

ADVANTAGES:

- Suitable if drug volume is moderate
- Suitable for certain oily / irritating substances
- Depot preparation can be injected by this route



DISADVANTAGES:

- Affects certain lab test (creatinine kinase)
- Can cause swelling at the injection site
- Can cause intramuscular hemorrhage (following anti-coagulant therapy)

SUBCUTANEOUS ROUTE

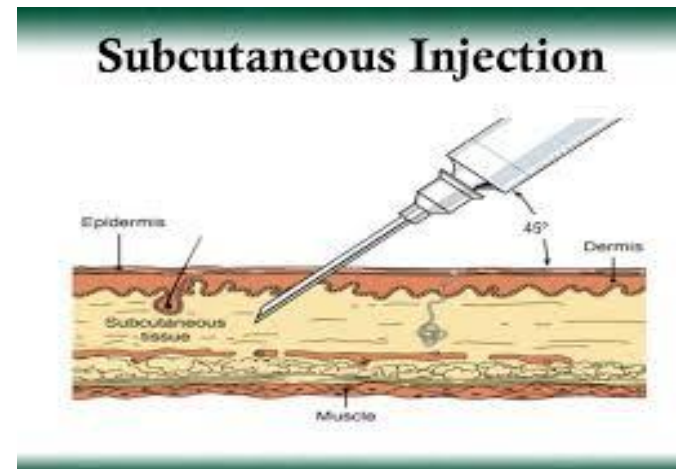
e.g. Subcutaneous insulin

ADVANTAGES:

- Drugs irritant to GI can be given
- Minimizes the risk of hemolysis or thrombosis associated with IV inj.
- Provides constant ,slow and sustained effects
- Self-administration is possible as deep penetration is not required

DISADVANTAGES:

- Pain or necrosis if the drug is irritating
- Unsuitable for the drugs administered in large volumes

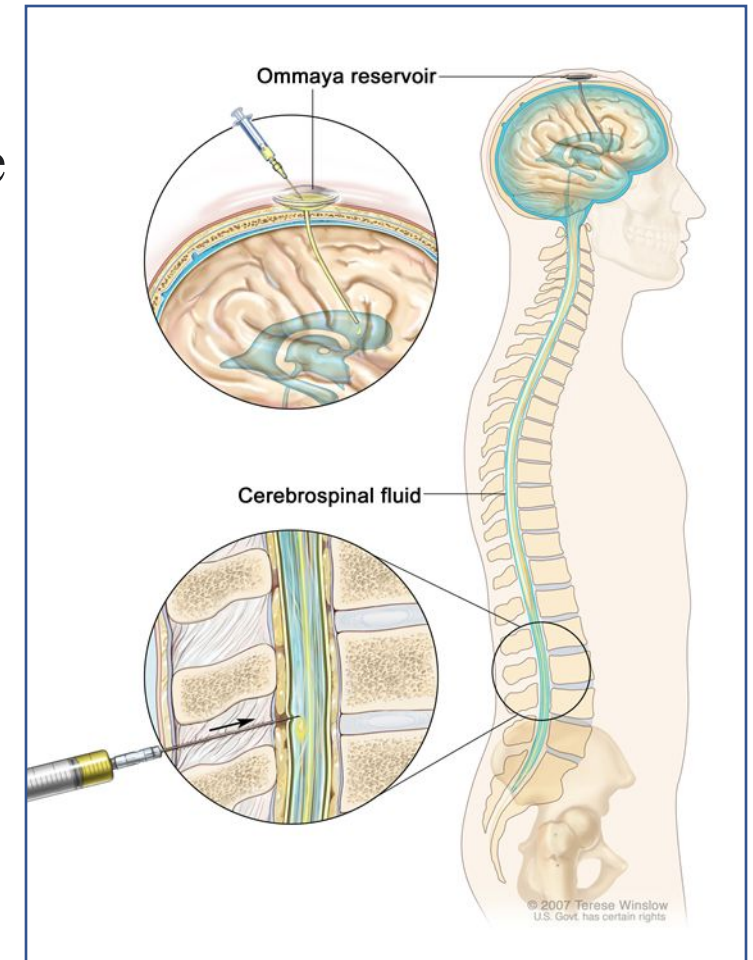


INTRATHECAL ROUTE:

Intrathecal is the route of administration for drugs via an injection into the subarachnoid space so that it reaches the cerebrospinal fluid (CSF)

APPLICATION:

1. Spinal anesthesia
2. Pain management

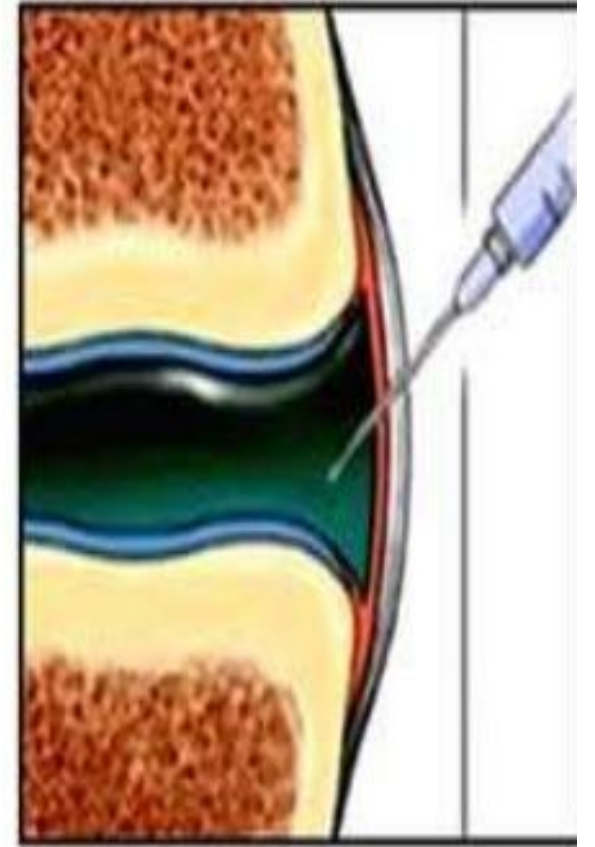


INTRA-ARTICULAR DRUG:

Intra-articular drug delivery is a method to administer drug into the joint cavity.

APPLICATION:

Inflammatory conditions (Osteoarthritis, RA)

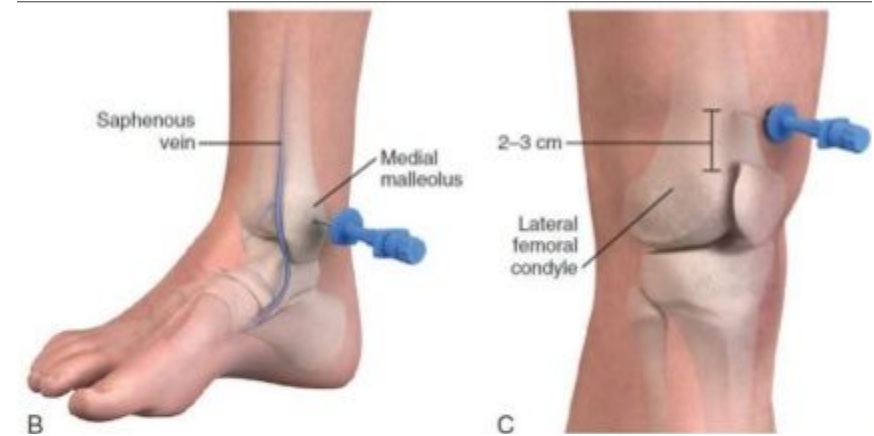


INTRA-OSSEOUS DRUG:

Intraosseous infusion (IO) is the process of injecting directly into the marrow of a bone

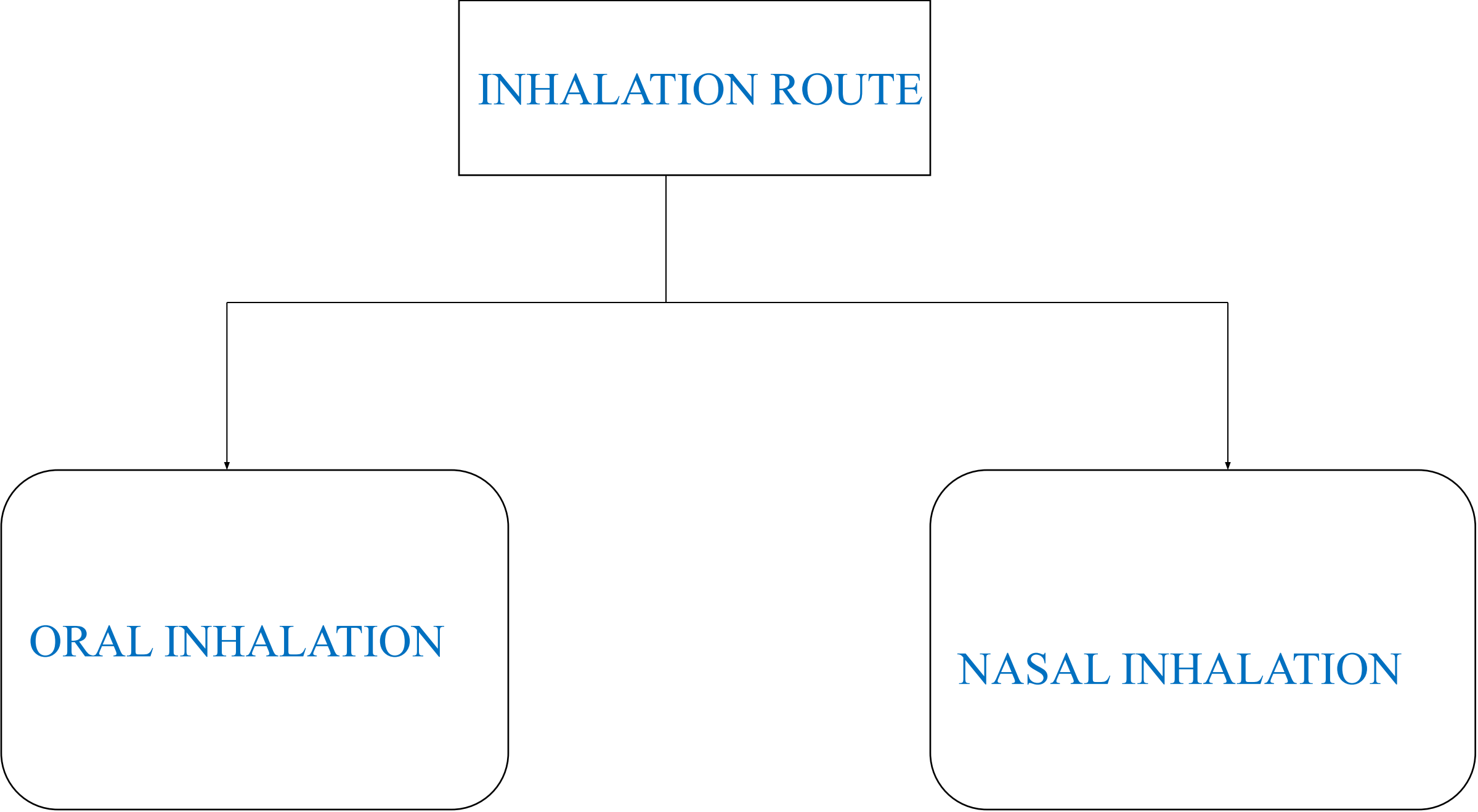
APPLICATION:

In children, in cases of emergency (shock, cardiac arrest)



INHALATIONAL ROUTE

INHALATION ROUTE



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graph TD; A[INHALATION ROUTE] --> B[ORAL INHALATION]; A --> C[NASAL INHALATION];
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A hierarchical flowchart starting with a rectangular box at the top labeled 'INHALATION ROUTE'. A vertical line descends from the center of this box to a horizontal line. From the left end of this horizontal line, a vertical line with a downward-pointing arrowhead leads to a rounded rectangular box labeled 'ORAL INHALATION'. From the right end of the horizontal line, another vertical line with a downward-pointing arrowhead leads to a rounded rectangular box labeled 'NASAL INHALATION'. All text is in a blue, serif, all-caps font.

ORAL INHALATION

NASAL INHALATION

INHALATION ROUTE

1. NASAL INHALATION

- Nasal-Decongestant e.g. (Corticosteroids / Oxymetazoline)



2. ORAL INHALATION

(e.g. bronchodilators, corticosteroids)



3. Nasal DROPS

(e.g. Intranasal drop of desmopressin in diabetes insipidus)



ADVANTAGES OF INHALATIONAL (ORAL AND NASAL):

- 1) Absorption is rapid, can have immediate effects
- 2) Ideal for gases (Inhalation anesthetics)
- 3) Effective for patients with respiratory problems (Asthma/COPD)
- 4) Fewer systemic side effects

DISADVANTAGES OF INHALATIONAL (ORAL AND NASAL):

- 1) Most addictive route (drug can enter the brain quickly)
- 2) Patient may have difficulty regulating the dose
- 3) Some patients may have difficulty using the inhalers

TRANSDERMAL ROUTE

e.g. Nicotine patches

This route of administration achieves systemic effects by application of the patch(containing drug) to the skin

ADVANTAGES:

- 1) Bypass the first pass effect
- 2) convenient to use / painless
- 3) Highly lipid soluble drugs e.g. estradiol/ nitroglycerin
- 4) absorption is slow /sustained /prolonged

DISADVANTAGES:

- 1) Some patients are allergic to patches that can cause irritation
- 2) may cause delayed delivery of the drug to the site of action



RECTAL/VAGINAL ROUTE

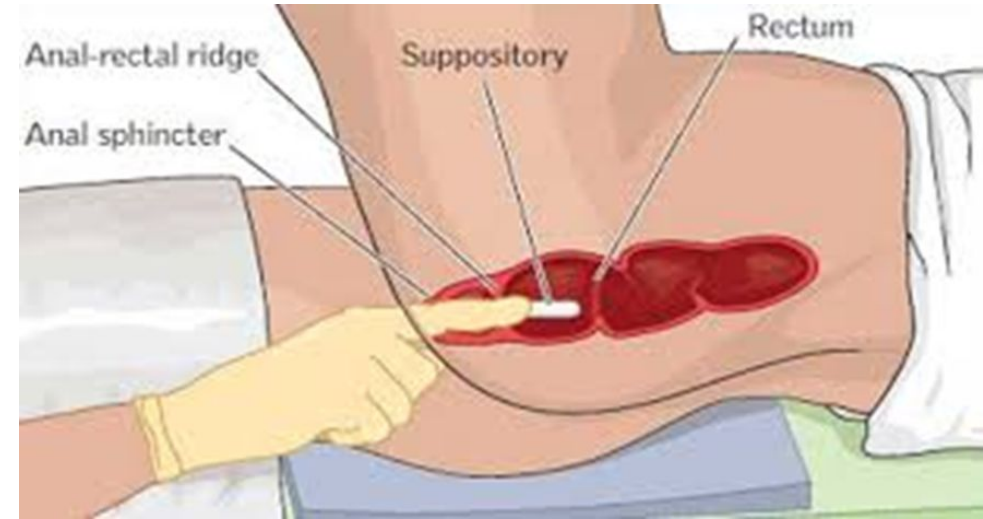
e.g. Glycerin suppository

ADVANTAGES

1. Partially bypass the first pass effect
2. Bypass the destruction by stomach acid
3. Ideal if drug causes vomiting
4. Ideal in patients who are vomiting or comatose pt.

DISADVANTAGES

1. Drugs may irritate the rectal mucosa
2. Not a well-accepted route



Parenteral administration

Advantages

- high bioavailability
- Rapid action (emergency)
- No first pass metabolism

Suitable for

- Vomiting & unconsciousness
- Irritant & Bad taste drugs.
- No gastric irritation
- No food-drug interaction

Dosage form:

Vial or ampoule

Disadvantages

- Infection
- Sterilization.
- Invasive assistance require
- Pain
- Needs skill
- Anaphylaxis
- Expensive.

Ampoule



iv infusion



Vial



*Thank
you*

